PRODUCT MANAGEMENT SYSTEM

Project Documentation & Reflection | Week 05 – Task 6

Student Name: Ankit Raj	**Project: Product Management System**
Organization: InternNova	Technology: HTML, CSS, JavaScript

1. Project Overview

The Product Management System is a simple, user-friendly web application developed to manage product information. It allows users to add products with their name, price and category, view product details, search the product list, filter products by category and delete products that are no longer needed. The interface uses clear sections and product cards to make information easy to understand.

2. Features Implemented

- Add a product by entering its name, price and category.
- Display product details in individual product cards.
- Delete products from the product list.
- Search products by name and filter them by category.
- Update the total product count dynamically.
- Validate required form fields before adding a product.

3. Technologies Used

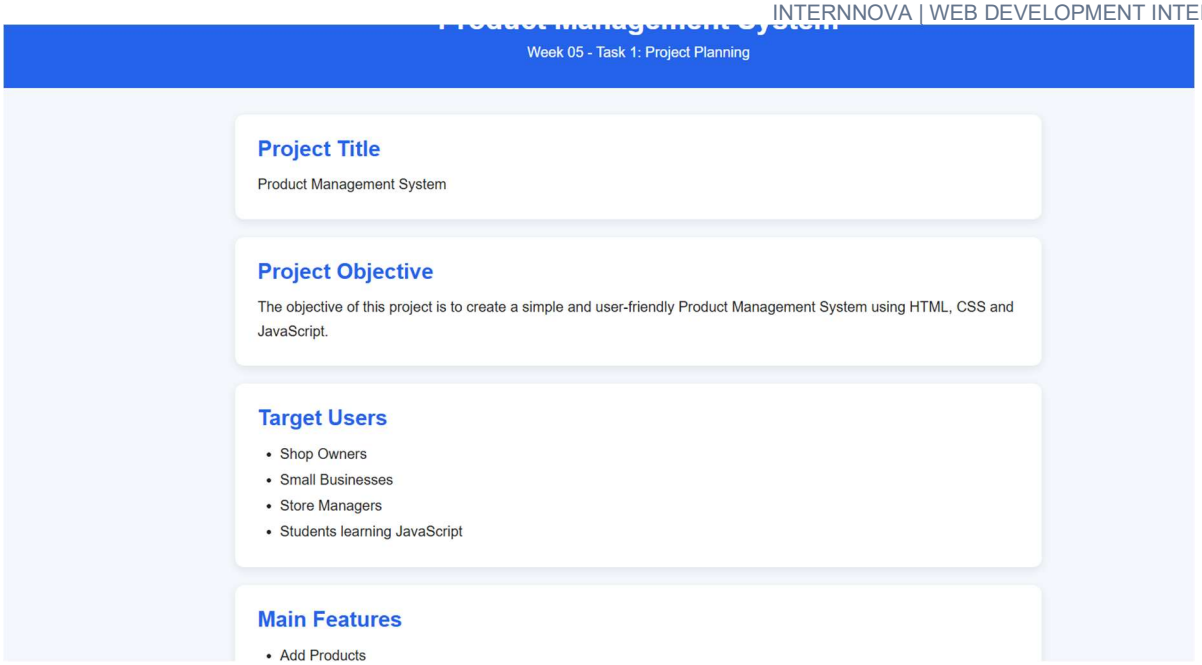
HTML5 was used to create the page structure and form elements. CSS3 was used for colors, spacing, cards, layout and responsive styling. JavaScript was used to implement product operations and make the interface interactive.

4. JavaScript Concepts Used

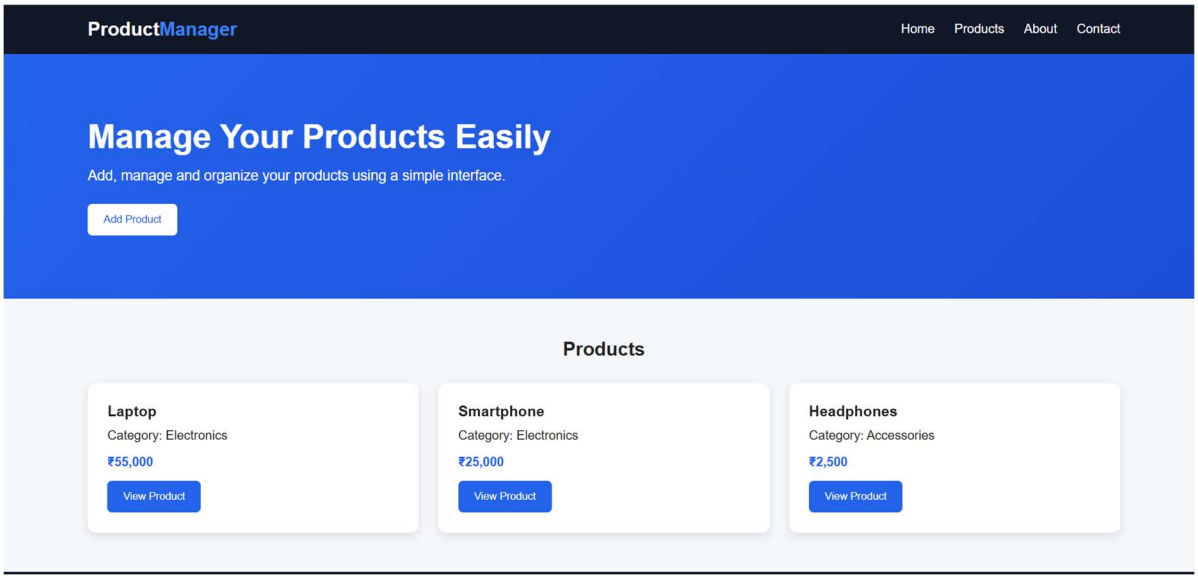
The project uses variables, functions, arrays and objects to store and manage product data. Conditional statements help validate input. Array methods such as push(), filter() and forEach() support adding, deleting, searching, filtering and displaying products. DOM selection and manipulation update the page, while event listeners respond to form submissions, button clicks and search input.

5. Screenshots of the Project

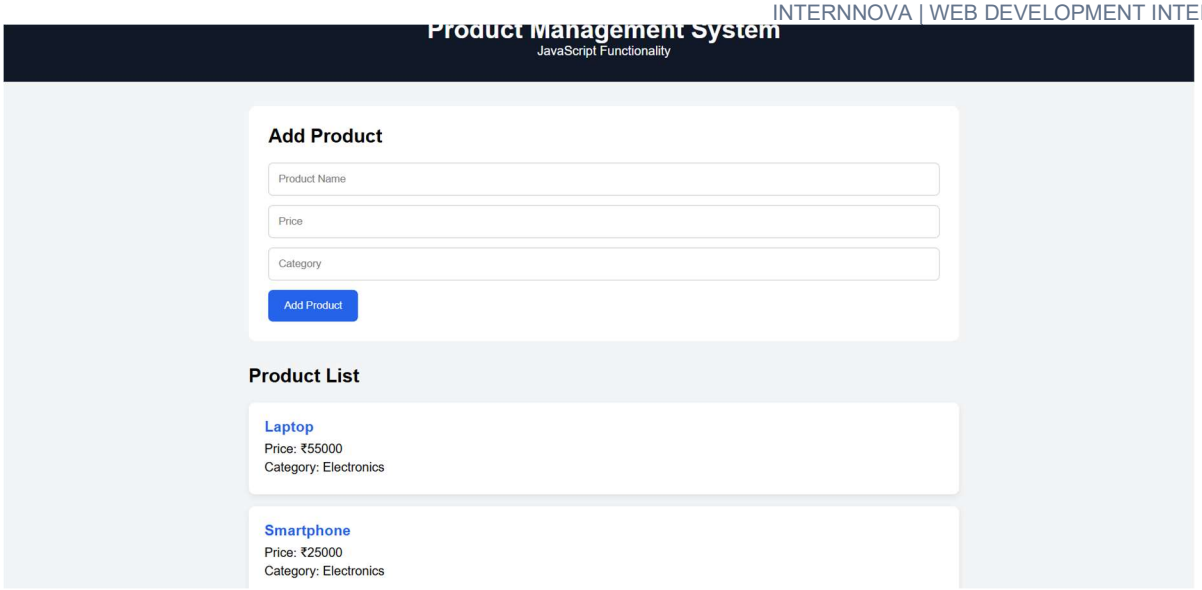
The screenshots below show the project planning, webpage structure, product functionality and final dashboard.



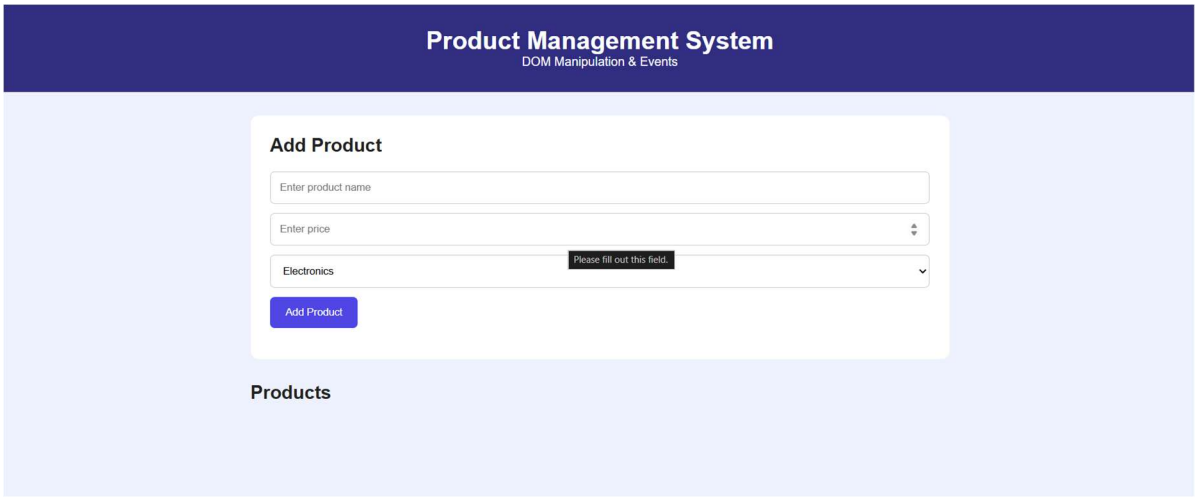
Screenshot 1 – Project planning: project title, objective, target users and planned features.



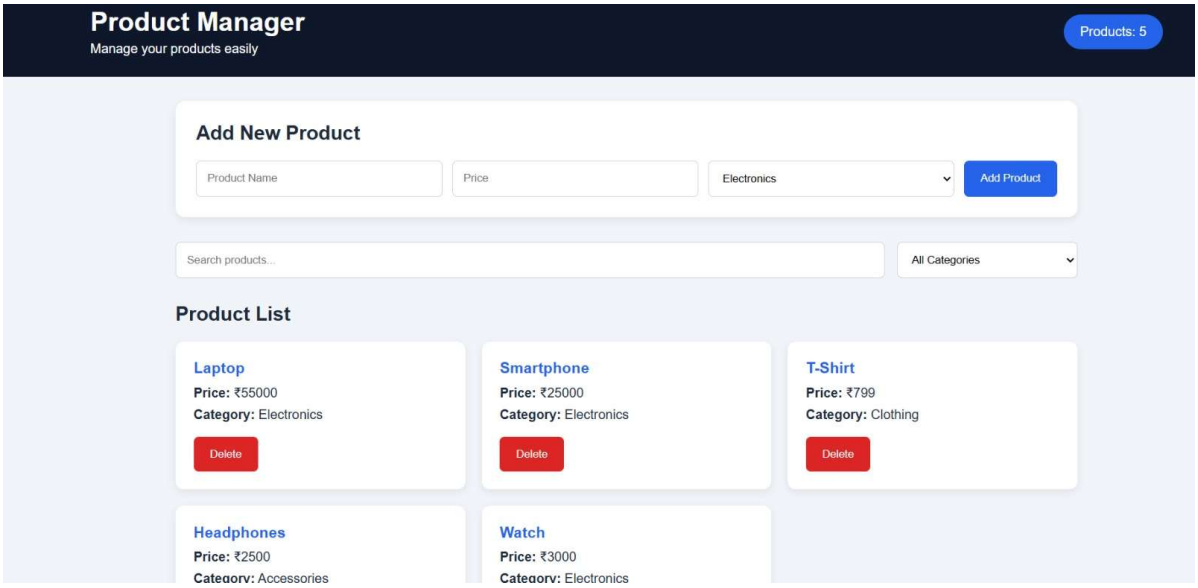
Screenshot 2 – Main webpage: navigation, hero section and product cards.



Screenshot 3 – JavaScript functionality: add-product form and product list.



Screenshot 4 – DOM manipulation and events: interactive product form.



Screenshot 5 – Product manager dashboard: add product, search, category filter and delete controls.

6. Challenges Faced

During development, the main challenges were displaying product data dynamically, validating form input and keeping the product count accurate after adding or deleting items. Combining search and category filtering also required careful handling of the product array.

7. Solutions Implemented

These challenges were addressed by separating the work into small JavaScript functions, checking input values before saving and re-rendering the product list after each change. Array filtering was used for search and category selection, and event listeners connected the controls to the required actions.

8. What I Learned

This project improved my understanding of how HTML, CSS and JavaScript work together. I practised handling user input, working with arrays and objects, using DOM methods and creating interactive features. I also learned the importance of testing each feature after making changes.

9. Future Improvements

Future versions could include editing product details, sorting by price, adding product images, saving data with localStorage and connecting the application to a database for permanent storage.